PETRA ZLATARIĆ-NOVOSEL

Machine Learning Engineer

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PROFILE

Machine Learning Engineer with 5+ years of experience across NLP, document classification, and production ML systems. Experienced in fine-tuning transformer models, serving models as FastAPI services in Docker on Azure, and monitoring production model quality. Strong Python and MLflow experience; PhD candidate researching Croatian language processing.

WORK EXPERIENCE

ML Engineer, Slavonica Document AI d.o.o., Osijek *03/2023 - Present*

- Developed and shipped a production document-classification system for OCR-processed incoming documents, including invoices, contracts, and decisions; achieved 94% accuracy and 0.91 macro F1 across 23 classes, processing approximately 15,000 documents per day.
- Fine-tuned XLM-R and BERTić transformer models with PyTorch and Hugging Face Transformers; served models as FastAPI services in Docker on Azure App Service, with Azure ML and MLflow for experiment tracking and model registry.
- Implemented production model-quality monitoring using Python, Evidently, Azure Functions, and PostgreSQL, tracking input text length, vocabulary, and predicted-class distribution; detected an OCR change that reduced accuracy by 9 percentage points and supported remediation within one week.
- Mentor two students on professional internships.

Data Scientist, Baranjska kreditna banka d.d., Zagreb (remote from Osijek) *09/2021 - 02/2023*

- Developed customer churn prediction models (gradient boosting, LightGBM) for the sales team.
- Developed customer complaint topic classification using a TF-IDF and logistic-regression baseline, followed by fastText, across 11 topics for weekly customer-service management reporting.
- Created SQL reports using an Oracle data warehouse.

Project Research Assistant, Faculty of Applied Informatics, Drava University, *10/2020 - 08/2021*

- Built a corpus and annotated Croatian legal texts; wrote Python scripts.

SELECTED PROJECTS

Retrieval-Augmented Generation (RAG) Question Answering Prototype | Personal project | 2025

- Built a weekend prototype for question answering over Croatian laws using Python, sentence-transformers, FAISS, and an open-weight LLM through an API. Prototype only; not used in production and evaluated through manual checks.

Croatian Named Entity Recognition | Slavonica Document AI + PhD | 2023–2025

- Built an annotated dataset of 4,000 Croatian legal and administrative documents using an annotation guideline and two annotators; achieved inter-annotator kappa of 0.87. Fine-tuned BERTić for persons, organisations, OIB numbers, addresses, and legal references.

EDUCATION

PhD studies in Computer Science (in progress, since 2022; expected 2027), Drava University - topic: named entity recognition in Croatian legal texts

MSc in Computer Science, Drava University, 2020

CERTIFICATIONS

Microsoft Certified: Azure AI Engineer Associate, 2024

SKILLS

Programming: Python, SQL
ML/NLP: PyTorch, Hugging Face Transformers, scikit-learn, LightGBM, fastText, spaCy
Serving and MLOps: FastAPI, Docker, MLflow, Azure ML, Azure App Service
Data: pandas, NumPy, Oracle SQL
Tools: Git, Linux

LANGUAGES

Croatian (native), English (C1), German (B2)

TEACHING

Guest Lecturer, NLP Summer School, 2025 — Delivered a 90-minute lecture on NLP.

PUBLICATIONS

- Zlatarić-Novosel, P. et al. (2024). Named entity recognition for Croatian administrative documents. Slavic Language Processing Workshop.